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77-68 USER GROUP
Fifth Newsletter THU 11 JAN 1979

Comment

Tim Moore tells me the score of 77/68 processor boards sold is now about 600 and that probably a hundred systems have been built on prototype or bread boards. Remarkably there has been very little of the criticism of the design that has been aimed at other systems. See, for example, R.A. Cottis's letter in the January ETI. The principle criticism of the various systems is mainly concerned with the difficulty of expansion as for some reason, probably cheapness, few systems have well buffered busses to the outside world. From the letters received this is an area where the 77/68 scores heavily.

Since the last news letter we have been innundated by a large amount of very useful material much of which is contained in this edition. A very exciting development is the 32K dynamic RAM board which should be available shortly from Newbear and will at last bring a decent size memory at an affordable cost. Another important development, although not specific to 77/68 is Andrew Holt's work on BASIC.

It is is very encouraging that interest in the association is turning to monitors and BASIC. It indicates that substantial systems are being built and used and I look forward to some exciting developments from members.

This quarter your worthy editor has built a MON 2 board and a Cottis Blandford CUTS interface both on boards from Newbear. Both projects presented few problems even though only preliminary documentation was available. In keeping with my other boards the MON2 connections to the outside world are brought out to metal brackets fixed to the end of the board away from the edge connector. This carries a V24 25 way connector and the single shot key. This technique has the advantage that any board can be put in any slot. Firstly, it is impossible to put a board in the wrong slot. Secondly it is easy to place a faulty or developing board in the end slot where it is easy to probe. The CUTS board has just enough room to mount V24 drivers and receivers to connect to the V24 connector on the machine. This also has the advantage of making it easy to connect to other computers in particular to make use of cross assemblers.

Murphy's law is dead! A gastly error caused the +12 volts and the -12 volts to be interchanged as well as the +5 and zero. The only damage was a broken V24 driver chip. There is more. My power

supply became unstabilised at 8 or 9 volts. Again no damage was done and the system even worked at that voltage although some chips got a bit warm. Dear reader you repeat the experiment at your own risk!

When I visited Newbear last week a chap brought in a superbly built MON2 board with every chip in a socket. I tend to use sockets sparingly. The advantages of sockets are that broken chips can be replaced easily or tested by replacement. In addition there is less danger of overheating chips during soldering. On the other hand sockets are expensive, often more expensive than the chips they hold. There are some good methods of removing soldered in chips. I have silver soldered an inch strip of 1/8 copper to a soldering iron bit which can then melt the solder on all the legs on one side so that the chip can be 'bent' off. I have found desoldering braid and suction devices useless. Chips, in fact, are quite difficult to damage by heat. My favourite way of stripping a board is to play a blow lamb on the back and flick the chips off. They get too hot to handle but they still work well.

The production of the diagrams in this newsletter is very time consuming. Since it is achieved by amateur effort it would be much appreciated if contributions could be on plain A4 paper and in black ink or pencil ready for reproduction. If anyone in the Newbury area would like to help with the production of the Newsletter please contact me. Proof readers and draftsmen are particularly welcome.

Money. Unfortunately this newsletter costs money to reproduce and post. To cover the costs (almost) a charge of about £1.50 a year will have to be made, details will be in the next newsletter.

Thank you for the many contributions and may I wish you all a belated happy New Year.

Paul Bryant
The Bumbles
Well Meadow
Shaw
Newbury

CORRECTIONS

Correction To MON 1 board

X28 on the MON 1 board is supposed to provide low baud rate clocks by dividing the 1200x16 Hz signal by any integer in the range

2-15. In fact, it will only work for division ratios of 8 or less if the output is taken from X28 pin 15, rather than X28 pin 11.

Correction To BUG 1 on Paper Tape

The following correction should be made:-

FFD6 should be 23B8	Corrected error
FF1B should be 8D65	Stops character
FFDC should be 8DA4	repeat

SOFTWARE

DEW FLASHER 2 PROG

Below is my 77/68 'DEW FLASHER 2 PROGRAM' it counts in binary until it matches SW reg. It then cycles all the lights on and off a certain number of times (stored in 80). Shorter programs were tried but did not work.

LOCATION	CONTENT	LOCATION	CONTENT	LOCATION	CONTENT
00	4F	0C	26	18	FD
01	D6	0D	F6	19	86
02	FF	0E	D6	1A	FF
03	5C	0F	80	1B	97
04	08	10	08	1C	FF
05	26	11	26	1D	5A
06	FD	12	FD	1E	26
07	97	13	4F	1F	F0
08	FF	14	97	20	01
09	4C	15	FF	21	01
0A	5A	16	08	22	20
0B	5D	17	26	23	DB
80	NUMBER OF ON/OFF CYCLES (EG 03)				

To run:- set switches to 00, hit reset, set sw. reg to required binary count. switch to 'run'.

Duncan Wills (age 13)

ED. Must be the youngest 77/68 constructor- unless you know different?

BASIC

6800 basic

There are at least 6 BASIC's available and probably many more.

Unfortunately they all need changes of one sort or another to make them work particularly with the 77/68 VDU. Several members have made relevant changes and some technical details are in the correspondence. The BASIC'S available are:-

2K byte. Tom Pitman's Tiny Basic. This is not available in the U.K. Try:- Itty Bitty Computer Co., PO Box 23189, San Jose, CA 95153, U.S.A. The documentation is excellent but the system only provides integer arithmetic.

3K byte. Micro BASIC. Computer Workshop/S.W.T.P. This has been adapted by Mick Alger to run on the 77/68, MON 1 and VDU board. The cassette is available at £2-50+p&p+VAT from Newbear but the manual is not (sorry about that) but can be obtained from Computer Workshop or S.W.T.P. at £5-00.

4K byte. Computer Workshop/S.W.T.P. This has floating point but is not recommended. It was published in a 1977 edition of Kilobaud. It is MIKBUG compatible and has been used on 77/68 by Tony Baldwin.

8K byte. Computer Workshop/S.W.T.P. Recommended for its low cost. It is MIKBUG compatible. It is not easy to modify its I/O routines to run under other monitors.

8K byte. Super Cassette Basic. Computerware. Loosely based on S.W.T.P. 8K BASIC but greatly improved and restructured for ease of use, speed (up 40%) and facilities. It has been used on 77/68. A new upgraded version is now available.

8K byte. PROM BASIC. Computerware. As above but can be used in PROM. Has been implemented on 77/68.

Tim Moore

CSS 'Super Cassette Basic' on a 77/68

Super Cassette Basic is a 9K BASIC for the 6800, similar to, but 50% faster than SWTP 8K BASIC. There are also some extra facilities, notably the ability to specify maximum string lengths (this requires a patch in SWTP V2.0), and the BASE=0 statement to allow arrays to have elements of subscript zero.

My 77/68 is configured to allow easy running of SWTP software, there are 128 bytes of RAM at A000, the usual MIKBUG addresses available in my monitor. Normally all I need to do to use software for a SWTP system is to patch in my 'break' routine as I use an ACIA for my teletype interface, and my peripheral addresses are all in the FOxx block.

CSS BASIC loaded OK from cassette once it had been discovered that the 'SWTBUG' format dump was in standard X1 format. The first thing to do was to patch in my 'break' routine so, looking at the I/O listing in the manual I planned the necessary patches. However the version supplied had rearranged the code and I had to search for the new locations of the break routine. The other problem I had initially was that the code to find out how much store was available does not work correctly on my 77/68. It goes up store storing a zero then reads back each location until a nonzero one is found. A better choice of value would have been \$55.

Once I found and corrected this the BASIC worked and I found the speed impressive, also the way that the line in error was printed beneath the error message.

There are some bugs in it, and some features which, while not bugs, cause some problems on a 'non-standard' system, but taken as a whole it is a considerable improvement on SWTP BASIC.

The problems with CSS BASIC V4.3 (and other oddities) are:-

- 1) Appendix F does not correspond with the supplied version.
- 2) The check for memory size does not work on a system where the bus drifts to \$00 and the code is too tight to be patched for a better check.
- 3) The maximum input line length is 82, not 123 as stated (80 is adequate so the document should be changed).
- 4) Changing the DELCHR from Ctrl/H causes problems with loading programs saved on SWTP BASIC, and it cannot be changed to the 'natural' character on a teletype (rub out) without further patches to the input routine.
- 5) Ctrl/D has a useful effect which is not documented. It would be even more useful if it did not do the final carriage return.
- 6) I have not yet had TLOAD successfully find a filename.
- 7) The following code crashes BASIC:-
100DEF FNA(X)=X+1
200FOR I=1 TO 40
300PRINT FNA(I),
400NEXT I

I could not even try the bug in SWTP BASIC which prevents nesting of user defined functions.

- 8) FOR I=4 TO 2 works as in SWTP BASIC (and in the manual), and not according to ANSI standard.
- 9) 9999END is rejected.
- 10) RND is biased about 6% high.
- 11) Appendix A is incorrect. LJUST, TRACE ON, TRACE OFF do not exist. Nor does OPTION (as mentioned on page 10).
- 12) No pointer to the error location is given (contrary to a statement in Appendix D).
- 13) The way most message strings are stored in memory does not allow them to be patched to shorter strings, yet there is no warning of this. The 'HOME' string is a prime example.
- 14) Ctrl/X causes 'DELETED' to be typed and cancels text as specified but it does no carriage return/line feed after the message, nor is there space to patch them in.
- I think it could be further speeded up by using a better symbol table format, and a better 'GOTO' method which does not search from the beginning of the program.

Andrew Holt

A later version of CSS BASIC is now available. Tim Moore.

SWTPC 3M MICROBASIC

Below are tested patches for SWTPC 3M MICROBASIC to run on 77/68 configured for MON 1.

ADDR	CONTENT	LABEL	CODE	COMMENT
003A	F07F			NEW X STACK TOP ADR.
0043	40			ADJUSTS MAX LINE LENGTH TO 64dec
0046	XXXX			ADR. OF LAST MEM LOCATION
				AVAILABLE IN YOUR SYSTEM
0294	7EFF7F		JMP PRNCM	NEW MONITOR ADR.
0297	7EFF7B		JMP GETCM	NEW MONITOR ADR.
029A	36	BREAK	PUSH A	THIS ROUTINE REPLACES 'BREAK'
	B6		LDA KBUFFZ	IN THE ORIGINAL PROGRAM.
	811B		CMPS 'ESC'	THIS CAN BE ANY SUITABLE CHAR
	2603		BNE BREAK2	
	7E0662		JMP 'READY'	
	32	BREAK2	PULL A	
	01		NOP	
	39		RTS	

```

0662 8EF045      LDS #$F045  PUTS STACK IN RAM ON CPU BOARD
066B 8EF045      LDS #$F045
066E CEF07F      LDX #$F07F  PUTS TOP OF X STACK IN CPU RAM
0694 8EF045      LDS #$F045
06CE CEF07F      LDX #$F07F
0730 BD0605 PATCH JSR NXTLIN
      33 CE07F5      LDX #$07F5  RETURN START ADR FOR PC
      36 FFF046      STX $F046   AT $F046
      39 8EF040      LDS #$F040  RETURN SP ADR AT $F008
      3C BFF008      STS $F008
      3F 7EFF1E      JMP MONITOR  OFF TO 77/68 BUG 2.

```

```

BEF008 PATCHRTN LDS $F008  CODE REQUIRED TO RETURN FROM
FEF046          LDX $F046  THE MONITOR WITHOUT RESTARTING
6E00           JMP 00,X   AND LOOSING YOUR PROGRAM.

```

```

021B 0D      C/R      THIS JUST ALTERS THE PRINT STRINGS
      1C 20      SP      TO CONTROL THE V.D.U.
      1D 20      SP      THE ORDER AND CONTENT MAY BE
      1E 20      SP      ALTERED TO SUIT YOUR TAST OR V.D.U
      1F 20      SP
0227 20      SP      AFTER PROMPT
022A 20      SP
      2B 20      SP
02C0 20      SP      CRLF STRING
      C1 20      SP
      C2 20      SP
      C3 0D
      C4 1E
      C5 20      SP      UNUSED
      C6 20      SP
      C7 20      SP
      C8 20      SP

```

Barry Benson

Prom Super Basic (Almost)

Some points of interest that may help others and a few points I would like help on. The PROM SUPER BASIC, being configured for MIKBUG, will not work with 77/68 and VDU, (memory map variety).

To get it to function:-

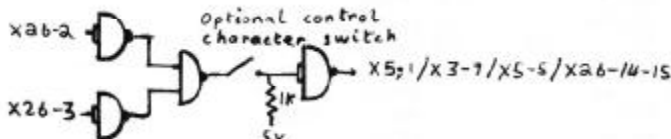
```

change address C01C to read 7EFF7F
      "          C01F      "  7EFF7B
      "          C103      "  201D

```

This results in output on the VDU.

Unfortunately control characters appear on the VDU. This can be cured by a modification to the VDU board shown below.



The pins on X5 and X26 fed in the above modification from the 74LS00 are grounded in the original design and tracks need cutting to make the mod.

BASIC will now function in some of its facilities:-

a) DIRECT PRINT

b) TSAVE, RUN, NEW etc.

It will not function on arithmetic commands and gives an error 16-MEMORY FULL. If the memory area allocated to 'MEMORY LIMIT' is examined it contains A7FE (since I have a 2K block of memory starting at A000 to satisfy the demands of PROM BASIC for 1/4K at A000 A0FF). If this is altered to contain 0FFF (end of my bottom block of memory) the BASIC will function on all direct commands but not on program listing when MEMORY FULL is still obtained.

That is the state of the art at present. I have not yet attempted dumping to cassette as I am unable to enter a program listing.

I would be glad of any suggestions regarding the 'ERROR 16'. I think that if I put 1/4K of memory at A000 to A0FF dedicated to PROM and move the 2K back to start at 1000 (continuation of my 4K initial block) things might be better.

I intend eventually to use a SMARTBUG and discard the memory mapped system as it seems to be the most compatible monitor. The only difficulty is that I cannot find anyone who offers this device for sale (at least not in any of the journals).

I would also like information on the BUG 2 NMI handling as the VDU book has me rather baffled.

David Woolf

Ed. Congratulation on getting so far with basic. Your problem smells very like Andrew Holts' problem with the code to determine the size of store. Can anyone help?. I will be pleased to pass on any information and also publish it in the next Newsletter.

HARDWARE

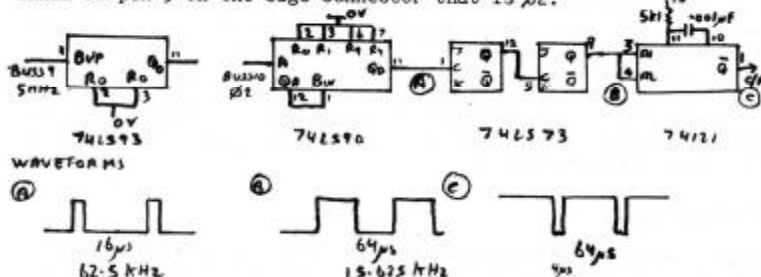
VDU 1 Board

On the next version of the 77/68 VDU 1 board please add an option so that the keyboard flag can trigger INT rather than NMI which is a disaster with most available software.

Andrew Holt

Memory Map VDU Picture Stability

Picture stability of the memory map VDU can be improved by deriving the 15.625 Khz from 02 or the 5 Mhz clock. The output from the divide chain, below, replaces the signal LS from X15 pin 5. For use with 02 74LS93 should be omitted and 74LS90 pin 14 taken to pin 9 on the edge connector that is 02.



Barry Benson

Memory Size Control (See Newsletter No. 4)

After the soothing editorial on P.C.B's I am forced to supply the following P.C.B. (see page 18). design for my memory size control system. One 7404 in the parts list should be altered to a 7402. The switch should connect +5V to SW64, SW32, SW16 or SW8.

The diagram is looking at the component side of the board. All ICs are 7402 except the two at the bottom which are 7404. All resistors are 4K7. Wire links are shown dotted. If 8K system is not required, use only the section above the dashed line and connect link *.

Why are there two tracks on the 77/68 board that go nowhere and connect to no device? Can you tell me what the DIL SPST switches given in the Newbear catalogue are?

Clive Feather

ED. I blush! Whilst I am indebted for the design I was really soliciting designs, schematic and possibly layout, to interface to the 77/68 bus.

The 8 inch square boards used in the 77/68 design are a 'standard' size or at least they appear in several catalogues, eg VERO. Thus the number of connections on the edge are Hobson's choice. The two spare connections are merely because there was no need to make any further connections via the connector. The 8 inch boards are a convenient size for this system as it does not cramp the design, which is important for a system expected to be tinkered with, and where breadboard designs are expected to be produced. The VERO catalogue has a number of prototype PCBs as well as plain boards with edge pads which can be etched to taste.

I am also often foxed by the description of components. My bete noir are the markings on diodes and capacitors which are either undecipherable or meaningless. DIL SPST switches are a set of 8 very miniature ones mounted in the same way as a Dual In Line 16 pin chip. SPST, I guess, means Single Pole Single Throw. The MON 2 board document recommends you to use them as handkeys for the processor board. OK as long as you do not bite your nails or have a suitable miniature finger.

77/68 VDU Board

Having recently completed construction of the 77/68 VDU board I thought you might be interested in my comment on Design Note 36.

Top of page 6:- Illustration of waveform is incorrect- X1 pin 6 waveform should be set to zero every seventh cycle of the waveform at X7 pin 8. X2 pin 12 blocks should be labelled 1,2,3,4,5,6,7.

Page 9:- 6.d) It is not clear why the system reset button needs to be held operated.

Page 9: 6.f) For completeness add reset vector to listing of test program.

Page 9: 6.g) The program as listed does not cause the display to continually shift. Due to the fact that the program accesses 768 VDU memory locations an exact multiple of the 256 displayed characters, a static display is obtained after the first scan of the display memory. Changing FFOE to 8CFB01 provides a 1 character offset to provide a shifting display.

Figure 1:- On initial testing of the P.C.B. I had a fault which caused only 12 rows of 40 characters to be displayed, This was

traced to a spurious pulse on the output of X1 pin 11 causing the 'row' counter to increment twice for every series of ten line pulses. X1 was generating a pulse at the start of line 8 as well as line 9. At the changover from line 7 to line 8 the following change occurs 1110 TO 0001. Since L0 changes from 1 to 0 and L3 changes from 0 to 1 this caused a spike a few nanoseconds wide to appear on X1 pin 11 and incorrectly update the row counter.

A 220pF capacitor was fitted between X1 pin 11 and 0 volts to cure the problem.

I also carried out the modification to shift the display up by one row as mentioned in User Group Newsletter No 4.

Figure 5:- Bottom right hand corner of memory map marked

FBA0-FBA7
FBC0-FBC7
FBE0-FBE7
should read
FBB0-FBB7
FBD0-FBD7
FBE0-FBE7

Figure 7:- Links A, B, and C should read A-B FOR UPPER & LOWER CASE and B-C FOR UPPER CASE ALPHA's ONLY

I hope these comments are of use to you.

L.T. Haynes

ADC/DAC/6522 Board

On page 19 there is a diagram of my ADC/DAC/6522 board for 77/68. I have deliberately not given 'noddy' type instructions as I suspect that people who did not have a TTL databook available should get someone else to do the job anyhow! Note that I use two 'non-standard' bus lines:- -5v and RESET (generated from my PROM board).

Bus buffers can be 81LS97 (best value) or 8212. The DACs can be ZN425E (my card) or MC1406. Connections to ZN425E are:-

data 0	pin 5
1	6
2	7
3	9
4	10
5	11
6	12
7	13

..V 1 & 2

5v 8

pins 1 & 2 are connected to 15 & 16 via a 0.22 F capacitor.

Address FF00-FF0F	is the 6522
FF10	write-set address and start ADC
FF10	read -ADC data
FF11	read -end of conversion flag
FF11	write-DAC1
FF12	write-DAC2

I have used this to effect in a moon landing game. Fuel rate is read from a pot. Height and speed are output to two meters. The code uses SWTP 8K BASIC.

My system is now:- CPU board, 4X4K RAM (0000-2FFF,A000-AFFF), 1X4K PROM (E000-EFFF), teletype, cassette, paper tape reader. Under construction at present:- IBM Selectric driver board (using SC/MP), improved cassette interface, VDU and Z80 processor board.

Andrew D. Holt

Ed. So we shall have to call ourselves 77/68 and 77/Z80 user Group.

MONITORS

77/68 VDU Driver Program

This program occupies 204 (decimal) or CC (hex) bytes. It is designed to drive the 77/68 VDU which has 24 rows of 40 characters each. As listed here it was extracted from my monitor program as it was at that time, which is why it starts at the unusual address of FC71. However it is completely relocatable.

The reason why such a long program is required to perform a comparatively simple function is that the VDU format of 24x40 is not completely divisible by binary numbers as would be 64x16 for example. Thus the 1024 byte RAM used to store the VDU characters can not be easily organised so that it can be accessed by the 10 address lines implied by a memory space of 1024 bytes. In fact the hardware is configured in such a way that the first 32 characters of the first 16 lines only are in the first 512 consecutive memory locations but after that it goes haywire.

The program has a central core section from FC95 to BCF2. It may be entered at FC95 with any 8 bit value in the A accumulator which

will be printed regardless of whether it is an alphanumeric or one of the rather odd graphics in the Texas 74S262 character generator ROM. The full program should be entered at the beginning which is FC71, in which case it will recognise a carriage return by printing spaces until the end of the current line and position the cursor at the beginning on the next.

Entry at either point is transparent to the calling program in that A B and X registers are all preserved for further use if required.

The cursor is non destructive. It merely inverts the characters previously at its location to show black on white. This was done so that the character can be recovered afterwards, for example in an on screen editing system which needs commands to move the cursor around.

Because of the complex memory map, the calculation of the next position on the screen involves rather more than just incrementing the X register as normally required in a simple VDU format. This is the function of the STEPLOC subroutine located from FC33 to FD3D. This particular routine has not been made transparent as there seemed no need, note that it returns with the new location in the X register as well as in VDULOC.

Apart from this description there are sufficient comments in the program itself to make details of its operation clear.

I am about 3/4 of the way through developing a monitor for 77/68 with the following facilities:-

My VDU driver and transparent 'home routine'.

On screen display of 276 bytes of memory with addresses and a cursor which is keyboard controlled to give full interactive editing of hex code.

Block move of any size.

Binary and MIKBUG load from tape.

Binary dump to tape.

Go to user program.

Software single step and trace with line by line display of op-code, registers, and next address.

Each of these modules has been tested individually already. It is just a question of joining them together and squeezing them into one K. In fact I have developed the software single step and trace on the basic 256 byte switches and light board only- that requires some patience!

00001	NAM	VDUPR
00002 FC71	ORG	\$FC71

00003	0020	SPACE	EQU	\$20	
00004	000D	CR	EQU	\$0D	
00005	F0FC	VDUHI	EQU	\$F0FC	
00006	F0FD	VDULOW	EQU	\$F0FD	
00007	F0F8	TEMPX2	EQU	\$F0F8	
00008	F0FC	VDULOC	EQU	\$F0FC	
00009	FC71 81 0D	OUTEE	CMP A	#CR	MIKBUG COMPATABLE
00010	FC73 26 20		BNE	PUTVDU	
00011	FC75 37		PSH B		
00012	FC76 36		PSH A		
00013	FC77 86 20		LDA A	#SPACE	
00014	FC79 F6 F0FC OUT1		LDA B	VDUHI	
00015	FC7C C1 FB		CMP B	#FB	
00016	FC7E 27 04		BEQ	OUT2	
00017	FC80 8D 13		BSR	PUTVDU	
00018	FC82 20 F5		BRA	OUT1	
00019	FC84 F6 F0FD OUT2		LDA B	VDULOW	
00020	FC87 53		COM B		
00021	FC88 C5 07		BIT B	#07	
00022	FC8A 27 04		BEQ	ENDOUT	
00023	FC8C 8D 07		BSR	PUTVDU	
00024	FC8E 20 F4		BRA	OUT2	
00025	FC90 8D 03	ENDOUT	BSR	PUTVDU	
00026	FC92 32		PUL A		
00027	FC93 33		PUL B		
00028	FC94 39		RTS		
00029	FC95 FF F0F8 PUTVDU		STX	TEMPX2	ENTRY FOR GRAPHICS
00030	FC98 FE F0FC		LDX	VDULOC	
00031	FC9B A7 00		STA A	0,X	PRINT CHARACTER
00032	FC9D 36		PSH A		
00033	FC9E 37		PSH B		
00034	FC9F 8C FBF7		CPX	#FBF7	END OF SCREEN?
00035	FCA2 27 04		BEQ	SCROLL	
00036	FCA4 8D 4D		BSR	STEPLOC	
00037	FCA6 20 3F		BRA	CURSOR	
00038	FCA8 CE F800 SCROLL		LDX	#F800	
00039	FCAB FF F0FC		STX	VDULOC	
00040	FCAE B6 F0FC SCR1		LDA A	VDUHI	
00041	FCB1 81 FB		CMP A	#FB	LAST FIFTH OF LINE?
00042	FCB3 26 14		BNE	SCR4	
00043	FCB5 B6 F0FD		LDA A	VDULOW	
00044	FCB8 43		COM A		
00045	FCB9 85 E0		BIT A	#E0	LINE 8 OR 16?
00046	FCBB 26 0C		BNE	SCR4	
00047	FCBD C6 D8		LDA B	#D8	
00048	FCBF 09	SCR5	DEX		
00049	FCC0 5A		DEC B		
00050	FCC1 26 FC		BNE	SCR5	
00051	FCC3 A6 00		LDA A	0,X	FROM LINE 9 OR 17

00052	FCC5	A7	D8		STA A	\$D8,X	TO LINE 8 OR 16
00053	FCC7	20	04		BRA	SCR6	
00054	FCC9	A6	20	SCR4	LDA A	#\$20,X	FROM SUCCEEDING LINE
00055	FCCB	A7	00		STA A	0,X	TO THIS LINE
00056	FCCD	8D	24	SCR6	BSR	STEPLOC	
ES057	FCCF	8C	FAEO		CPX	#\$FAEO	FILL LAST LINE WITH SPAC
00058	FCD2	26	DA		BNE	SCR1	
00059	FCD4	86	20	SCR2	LDA A	#SPACE	
00060	FCD6	A7	00		STA A	0,X	
00061	FCD8	8C	FBF7		CPX	#\$FBF7	END OF LINE
00062	FCD8	27	04		BEQ	SCR3	
00063	FCD8	8D	14		BSR	STEPLOC	
00064	FCD8	20	F3		BRA	SCR2	
00065	FCE1	CE	FAEO	SCR3	LDX	#\$FAEO	
00066	FCE4	FF	F0FC		STX	VDULOC	BEGINNING OF LAST LINE
00067	FCE7	A6	00	CURSOR	LDA A	0,X	TRANSPARENT CURSOR
00068	FCE9	8A	80		ORA A	#	
00069	FCEB	A7	00		STA A	0,X	
00070	FCEB	33			PUL B		
00071	FCEE	32			PUL A		
00072	FCEF	FE	F0F8		LDX	TEMP12	
00073	FCF2	39			RTS		END OF OUTEEE,PUTYDU
00074	FCF3	B6	F0FD	STEPLO	LDA A	VDULOW	CALCULATES NEXT POSITION
00075	FCF6	16			TAB		
00076	FCF7	43			COM A		
00077	FCF8	85	1F		BIT A	#\$1F	32ND COLUMN?
00078	FCFA	26	15		BNE	STEP1	
00079	FCFC	B6	F0FC		LDA A	VDUHI	
00080	FCFF	84	0F		AND A	#\$0F	=08,09 OR 0A
00081	FD01	48			ASL A		=08,09 OR 0A
00082	FD02	48			ASL A		=08,09 OR 0A
00083	FD03	48			ASL A		=08,09 OR 0A
00084	FD04	1B			ABA		
00085	FD05	8B	A1		ADD A	#\$A1	
00086	FD07	B7	F0FD		STA A	VDULOW	
00087	FD0A	86	FB		LDA A	#\$FB	
00088	FD0C	B7	F0FC		STA A	VDUHI	
00089	FD0F	20	29		BRA	STEP2	
00090	FD11	B6	F0FC	STEP1	LDA A	VDUHI	
00091	FD14	81	FB		CMP A	#\$FB	COLUMN 33 TO 40?
00092	FD16	26	1F		BNE	STEP3	
00093	FD18	17			TBA		
00094	FD19	53			COM B		
00095	FD1A	C5	07		BIT B	#\$07	END OF LINE?
00096	FD1C	26	19		BNE	STEP3	
00097	FD1E	53			COM B		RESTORE B
00098	FD1F	C4	18		AND B	#\$18	IF 00,ADD FD19;08,ADD FE11;
00099	FD21	10			SBA		10,ADD FFO9
00100	FD22	8B	19		ADD A	#\$19	

00101	FD24 B7 FOFD	STA A	VDULOW	
00102	FD27 07	TPA		SAVE CARRY
00103	FD28 54	LSR B		
00104	FD29 54	LSR B		
00105	FD2A 54	LSR B		0,1 OR 2
00106	FD2B 06	TAP		RECOVER CARRY
00107	FD2C B6 FOFD	LDA A	VDUHI	
00108	FD2F 89 FD	ADC A	##FD	
00109	FD31 1B	ABA		
00110	FD32 B7 FOFD	STA A	VDUHI	
00111	FD35 20 03	BRA	STEP2	
00112	FD37 7C FOFD STEP3	INC	VDULOW	
00113	FD3A FE FOFD STEP2	LDX	VDULOC	NEW POSITION
00114	FD3D 39	RTS		END OF STEPLOW
00115		END		

SYMBOL TABLE

SPACE	0020	CR	000D	VDUHI	FOFC	VDULOW	FOFD	TEMPX2	FOF8
VDULOC	FOFC	OUTEEE	FC71	OUT1	FC79	OUT2	FC84	ENDOUT	FC90
PUTVDU	FC95	SCROLL	FCA8	SCR1	FCAE	SCR5	PCBF	SCR4	FCC9
SCR6	FCCD	SCR2	FCD4	SCR3	FCE1	CURSOR	FCE7	STEPLO	FCF3
STEP1	FD11	STEP3	FD37	STEP2	FD3A				
S0060000484521B									
S112FC71810D262037368620F6F0FCC1FB2704D0									
S113FC808D1320F5F6F0FD53C50727048D0720F4E6									
S113FC908D03323339FFFOF8FEF0FCA70036378CC1									
S113FCA0FBF727048D4D203FCEFB00FFF0FCB6FOA3									
S113FCB0FC81FB2614B6F0FD4385E0260CC6D8096A									
S113FCC05A26FCA600A7D82004A620A7008D248CC1									
S113FCD0FAE026DA8620A7008CFBF727048D14208F									
S113FCF0F3CEFAE0FFF0FCA6008A80A7003332FED0									
S113FCF0F0F839B6F0FD1643851F2615B6F0FC84DE									
S113FD000F4848481B8BA1B7F0FD86FBB7F0FC20D9									
S113FD1029B6F0FC81FB261F1753C507261953C4C7									
S113FD2018108B19B7F0FD0754545406B6F0FC892B									
S111FD30FD1BB7F0FC20037CF0FDFEF0FC3957									
S9030000FC									

John Moore

T-BUG

Here is a brief preview of my 1K monitor 'T-BUG' for the 77/68 system soon to be available in PROM from the Newbear stall if there is enough demand.

It reads and accepts commands from any of three ports two parallel and one serial (intended for keyboard, papertape reader and cassette/teletype respectively). T-BUG outputs to the serial port

and the 77/68 VDU providing a scrolling display of 40 X 24 with cursor. T-BUG contains MIKBUG compatible routines at their proper addresses (no more patching of commercial software!). Other routines are also available to the user. Of particular interest is the plot routine: two bytes in RAM specify coordinates (00,00=top left and 27,17=bottom right corners of the screen) ACC B contains the character to be displayed, jumping to sub routine PLOT causes this character to appear in the specified location on the screen.

T-BUG commands enable the following:- memory inspect/modify, execute RTI instruction, print stack, dump contents of a memory block to cassette in standard MIKBUG compatible 'S!' format, load data in S! format into memory, block move, calculate branch offset, change serial data format (1 or 2 stop bits), display a table of the contents of a block of memory on the VDU, jump to location N, cassette file handling which enables named dumps and loads to/from cassette to be made and a list of the names of all the files on cassette to be displayed. T-BUG will live on the MON 2 board with its associated P.I.A. and A.C.I.A. If you are confused by all this be patient! A design note explaining T-BUG in detail will appear soon if there is sufficient interest.

Terry Cassell

MEMORY SIZE CONTROL

